Design of a Temperature-Compensated PLL Frequency Synthesiser

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Guide's Signature with Date

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Date: _____

Introduction & Problem Recap

- **Problem recap**
  - A PLL's ring-VCO drifts with temperature -- threshold voltage, mobility and capacitance all shift, so the same control voltage gives a different frequency hot vs. cold.
  - A compensation network cancels this by design, but sizing it needs a drift figure measured on the real topology, in the real process -- not assumed.
- **Objectives**
  - Design an integer-N charge-pump PLL (ref -> PFD -> charge pump -> loop filter -> 5-stage current-starved ring VCO -> /8 divider) in sky130, open-source, headless.
  - Measure uncompensated drift across all 5 process corners, then design and validate a temperature-compensation network with real statistical rigor (Monte Carlo + hypothesis testing).
- **Refinements after Review-I feedback**
  - Full 5-corner grid run (tt/ff/ss/sf/fs), not a 3-corner spot-check -- corrected the reported tempco spread from 3.4x to the real 12.5x (112-1399 ppm/C).
  - Cold-start acquisition risk (flagged as open in Review 1) fixed and verified: a true 0V cold start now locks in 839 ns.
  - Added a second-order (curvature-matched) compensation network, novelty-checked against prior art before committing to the claim (see Literature Review).

Detailed Literature Review

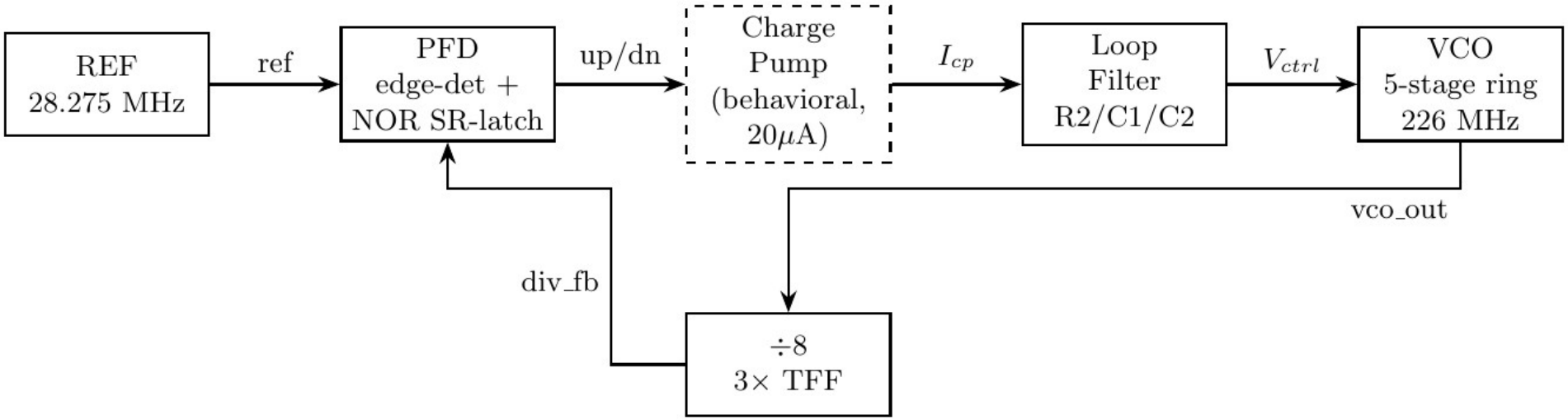
No.	Author(s) & Year	Title / Source	Method / Approach	Findings & Research Gap
1	G. Souliotis et al., 2022	Temperature compensated ring oscillator based VCO, AEU Int. J. Electron. Commun.	PTAT current compensation of a ring VCO inside a PLL; 180nm, 480MHz, fabricated	Closest topology benchmark: -1.5%/+0.4% over -40->120C, measured. First-order only; different node
2	C. Stancu et al., 2023	Temp. and power-supply compensated CMOS clock circuit based on ring osc., Electronics	Temp + supply compensated 50MHz ring-oscillator clock; simulation only	1.6% temperature error pre-layout. A clock circuit, never closed into a PLL
3	Z. Geng et al., 2018	Low power fast locking PLL freq. synthesizer with temp. compensation, EAME	Freq. presetting + dynamic loop BW + static/dynamic temp. comp.; 180nm, 2.1-2.6GHz	Lock <2us over 0-80C, -117dBc/Hz@1MHz, 9mW. Stability claim is the locked output, set by the crystal
4	Y. You et al., 2014	12GHz 67% tuning, 0.37ps RJrms PLL, LC-VCO temp. comp., 0.13um CMOS, IEEE RFIC	LC-VCO PLL for serial links; comp. scheme preserves tuning range; 5.6-13.4GHz	Stays locked -40->85C. Needs inductor models our process supports poorly; not portable to a ring VCO
5	S. Sinha, 2002	Design of an integrated CMOS PLL frequency synthesizer, IEEE MELECON	Dual-loop CP-PLL, 0.35um, 2.4-2.5GHz, 1MHz steps	-82dBc/Hz@10kHz. Architectural reference for our loop; no temperature compensation
6	E. M. V. Freitas et al., 2025	Using open source EDA tools in ASICs for HEP: a mixed comparison, JINST	Open-source vs commercial flows on IHP 130nm blocks incl. a current-starved ring VCO	Digital QoR quantified (+53% area, +118% power); analog VCO comparison qualitative, open-loop, no MC
7	M. Thiriloganathan et al., 2026	2.4GHz LC-VCO fractional-N PLL, open-source, 130nm BiCMOS, SMACD	Fully open-source flow (Xschem/ngspice/KLayout/OpenEMS), EM-modelled inductor	-100.8dBc/Hz@1MHz. Closes a loop open-source, but no temp. compensation, no PVT/Monte Carlo
8	R. N. S. Raphael et al., 2019	PLL-based frequency synthesizer for digital systems driving, BTSym'18 (Springer)	3rd-order CP-PLL for clock generation, integer-N (N=8-16), VCO input-cap-aware LF model	Settling <2us. Same loop class as ours; no temperature compensation

Detailed Literature Review (contd.)

No.	Author(s) & Year	Title / Source	Method / Approach	Findings & Research Gap
9	Y. Li, K. Du, J. Zhang, 2022	Design of ring oscillator with temperature compensation effect, Int. J. RF Microw. CAE	Positive-TC supply voltage on a 31-stage ring, 0.18um; 33kHz sleep timer	120 ppm/C (1.6%) over -40->80C. Single linear coefficient; kHz timer, not a PLL
10	H. H. Hammam et al., 2022	An ultra-low-power process-and-temperature compensated ring oscillator, IEEE ICEEE	3-stage current-starved ring, 130nm, process + temperature compensation	Up to 90% less variation vs. plain current-starved ring, 346nW. Standalone oscillator
11	S. Zhang et al., 2016	Temperature compensation technique for ring oscillators with tail current, Electron. Lett.	Tail current = constant source + PTAT current, first-order	Foundational first-order technique; no numbers reported in abstract; not closed-loop
12	M. Ortner, M. Punzenberger, 2017	Oscillators using quadratic temperature compensation, US Pat. 9,748,962 B2	Linear + quadratic temperature term for a VCO, -40->85C	Generic 2nd-order compensation is patented -- narrows our claim to ring-VCO + closed PLL + open PDK + stats
13	D. Patrinos, G. Souliotis, 2025	Low-voltage, low-power 2.5GHz ring oscillator with process & temp. comp., JLPEA	PTAT current + process-corner detection/tuning; 22nm, 0.8V; post-layout sim	Drift 23.9%->5.4% (tt) and 47.3%->7.2% (PVT). Corner-aware like our C(corner); not in a PLL
14	M. Sivasakthi, P. Radhika, 2024	PVT tolerant hybrid current starved ring VCO, bulk-driven keeper, 45nm, AEU	Two current-starved ring VCOs; Cadence GPDK45; corner + Monte Carlo	6.97GHz, -121.5dBc/Hz@1MHz. PVT tolerance of the VCO itself, no compensation network or loop
15	S. Madheswaran, R. Panneerselvam, 2024	7-stage MCML power-gated current starved VCO for PLL, IJECE	MOS current-mode logic delay cells with power gating, 45nm, simulation	3.06GHz, -125.9dBc/Hz. Same device class as ours; no closed-loop temperature characterisation

- Research gap (from all 15): no reported work combines a compensated ring VCO inside a closed loop, characterised across process corners with Monte Carlo rigor -- that gap is what this project fills.

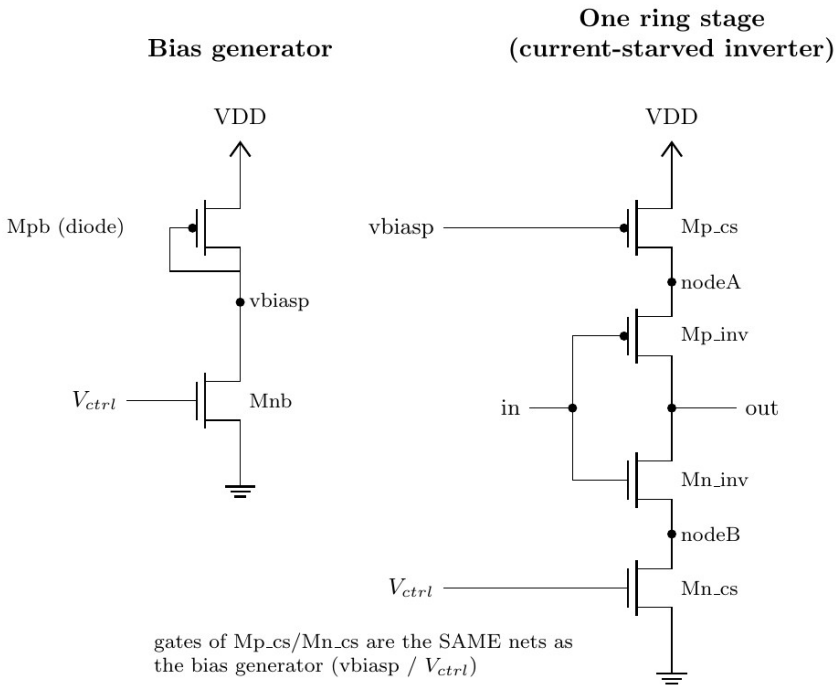
System Design & Architecture

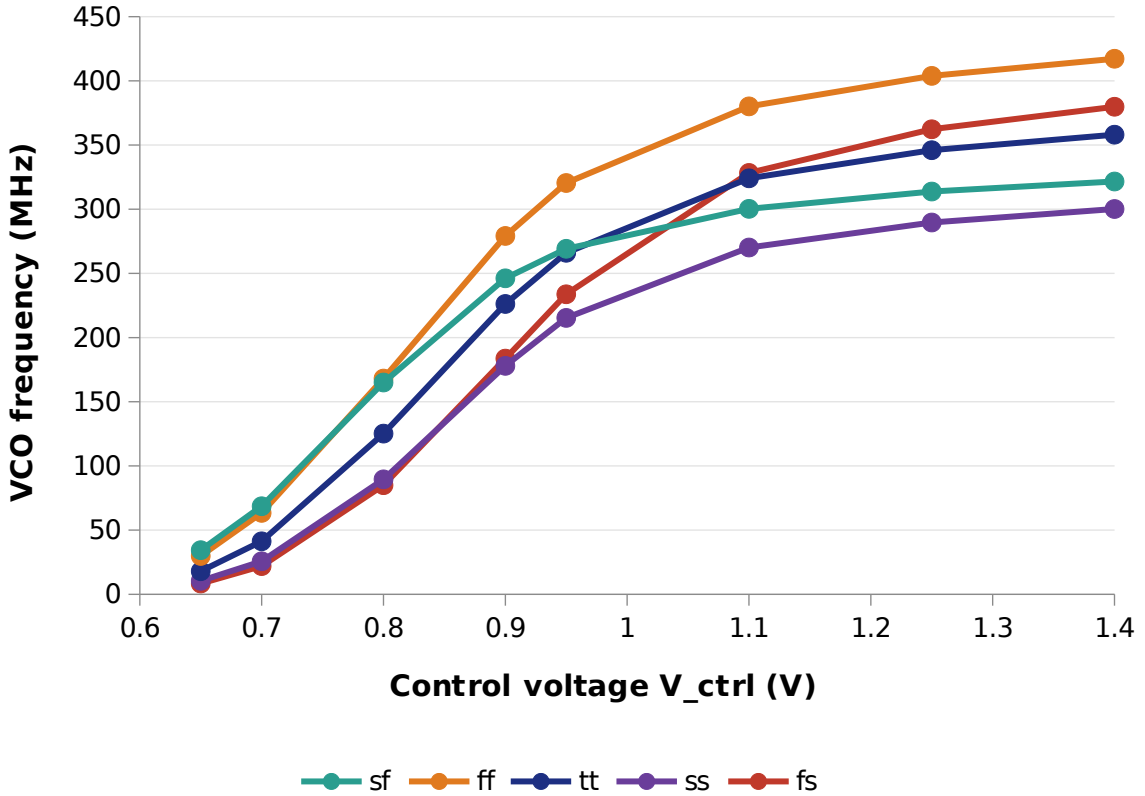


Closed-loop lock (tt, 27°C): **577 ns** (spec <2 μs, analytical model 450 ns)

- Integer-N charge-pump PLL: REF -> PFD -> charge pump -> passive loop filter -> 5-stage current-starved ring VCO -> /8 divider, with a compensation network on the VCO bias path.
- Scoping decisions (deliberate, documented): charge pump is an idealised 20uA behavioural source (transistor-level is Review 3, once layout needs real devices anyway).
- PFD uses edge-detectors + NOR SR-latches, not resettable DFFs -- a reliable async reset fights the latch's own bistable feedback loop.

VCO: Schematic & Tuning Curve





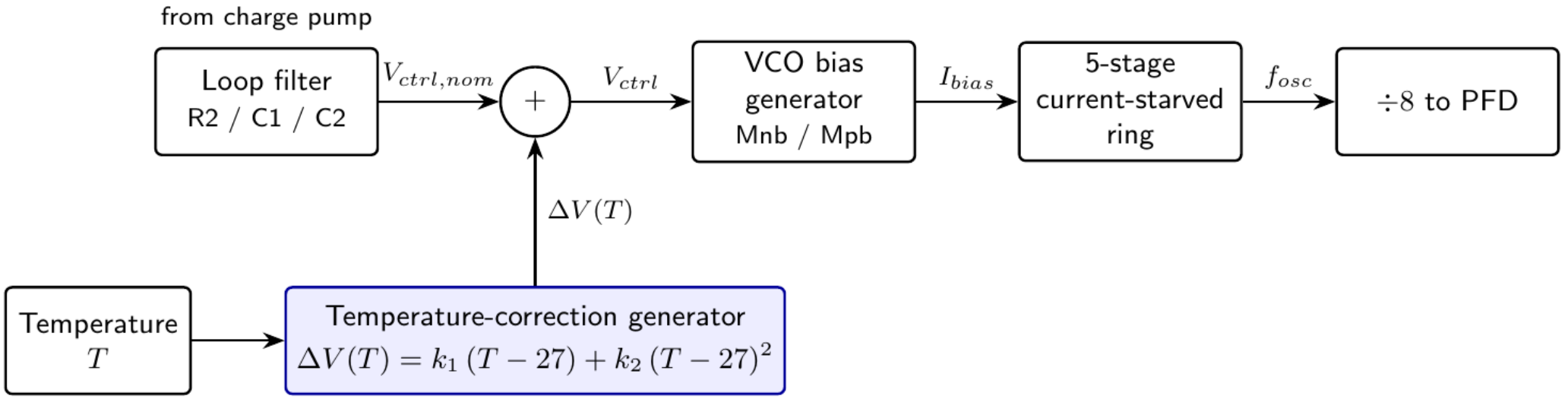
669 MHz/V
K_VCO at 0.9 V (tt)

≈ 0.63 V
oscillation floor

5-stage current-starved ring: V_ctrl sets the starving current, so it sets the delay per stage.

Design point 0.9 V gives 226 MHz = 8 × 28.275 MHz reference.

Temperature-Compensation Architecture



1st-order: $k_2 = 0$ (linear, PTAT/CTAT-equivalent) **2nd-order:** $k_2 \neq 0$ (curvature-matched)
Coefficients fitted per process corner; an ideal behavioural source in Review 2.

ΔV(T)
correction added to V_ctrl before the VCO bias generator

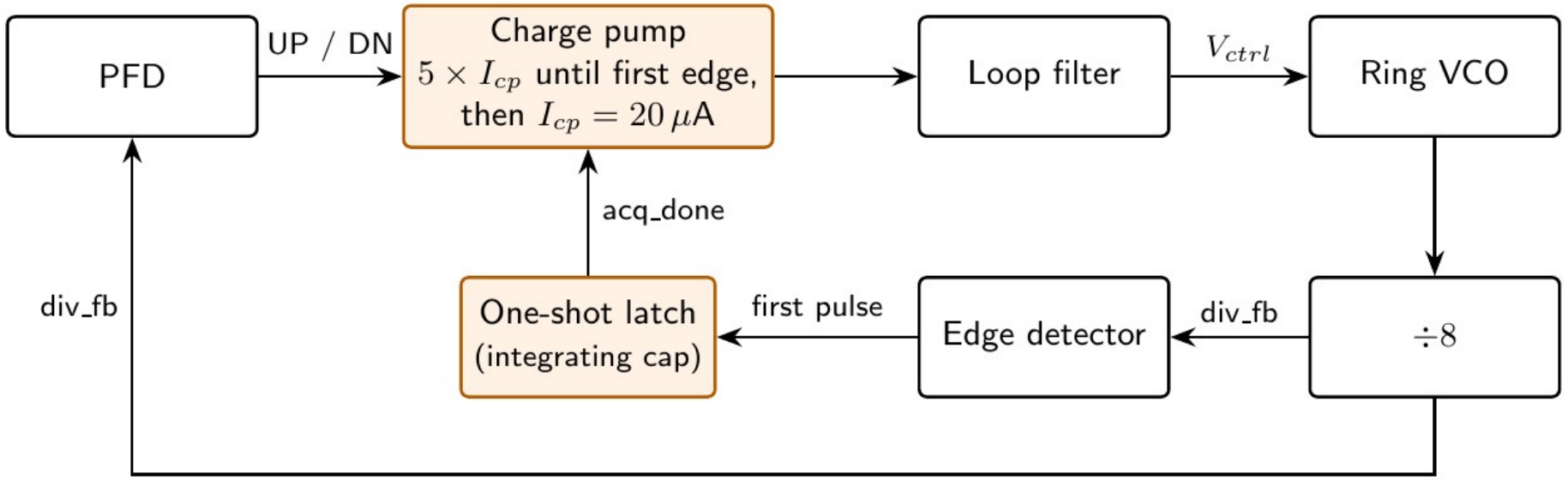
1st-order
linear k1·(T−27): PTAT/CTAT equivalent

2nd-order
adds k2·(T−27)²: matches the drift's curvature

per corner
k1, k2 fitted separately at tt/ff/ss/sf/fs

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Cold-Start Fix: Boosted Charge Pump



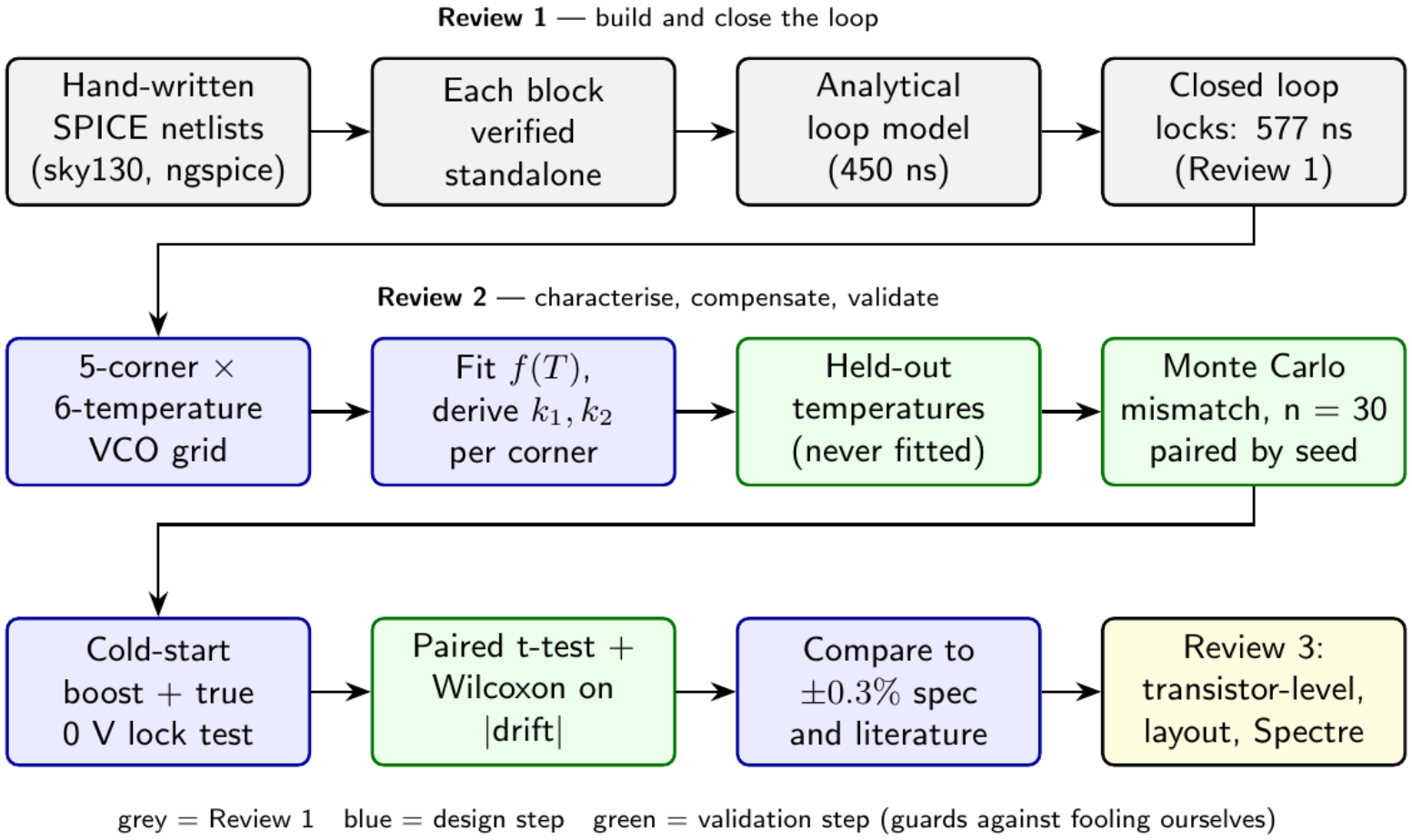
Problem (open since Review 1)

From a true 0 V start, V_ctrl sits below the 0.63 V oscillation floor: no VCO edges → divider silent → PFD sees only REF → slow pump-up.

Fix

Charge pump runs at 5 × I_cp until the divider's first edge, then a one-shot latch drops it to the nominal 20 µA. Result: lock in 839 ns from 0 V (< 2 µs spec).

Methodology: Review 1 → Review 2 → Review 3



Implementation Details

- **Modules developed and integrated**
  - VCO (5-stage current-starved ring), PFD, charge pump, passive loop filter, /8 divider, digital primitive cell library (INV/TG/DFF/TFF) -- all hand-written SPICE, ngspice, headless.
  - Temperature-compensation network: 1st-order (PTAT/CTAT-equivalent) and empirically-refined 2nd-order (curvature-matched), as a parameterised behavioural correction voltage.
  - Acquisition-boosted charge pump (5x current until the divider's first edge) -- closes the cold-start dead-zone risk carried from Review 1.
- **Algorithms implemented**
  - Coefficient derivation: analytical 1st-order (from the measured f(T) polynomial fit and K_VCO), then empirical 2nd-order refit against the measured post-1st-order residual -- zero extra simulation for the refinement step.
  - Statistics: paired t-test, Wilcoxon signed-rank (normal approximation), OLS regression with slope confidence interval -- hand-implemented (no scipy in this environment), self-tested.
- **Code snippet -- the compensation correction (ngspice behavioural voltage source)**

```
Bcomp vctrl vctrl_nom V = 'k1*(T-27) + k2*(T-27)*(T-27)'
  ; k1 = -1.322e-4 V/C, k2 = 2.891e-7 V/C^2 (tt, empirically refined)
  ; verified via an isolated 4-line polarity test before use in the full loop
```

Results & Analysis (75%)

- VCO tuning: K_VCO ~= 669 MHz/V at 0.9V (polynomial fit, not a secant); 18->358 MHz over 0.65->1.40V. Oscillation lost below ~0.60-0.65V (device-threshold-limited).
- Full 5-corner grid (all of tt/ff/ss/sf/fs, not a 3-corner spot-check) -- corrects the Review-1 finding: real tempco spread is 12.5x, not 3.4x. Worst corner (fs) wasn't checked before.

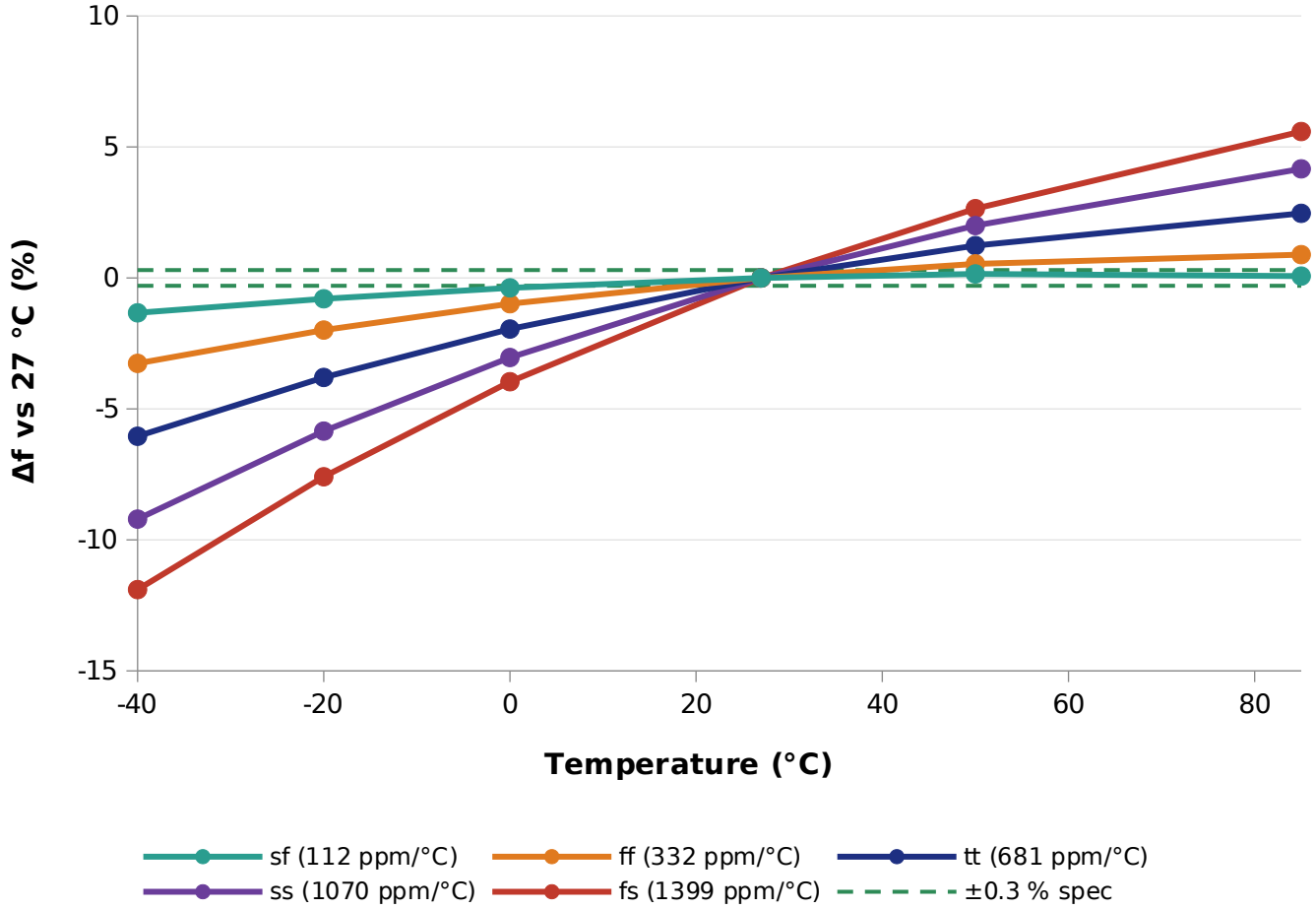
Corner	NMOS/PMOS	K_VCO (MHz/V)	Uncompensated tempco
sf	slow / fast	580.2	+112 ppm/C (flattest)
ff	fast / fast	763.6	+332 ppm/C
tt	typical / typical	669.3	+681 ppm/C
ss	slow / slow	568.2	+1070 ppm/C
fs	fast / slow	688.7	+1399 ppm/C (steepest)

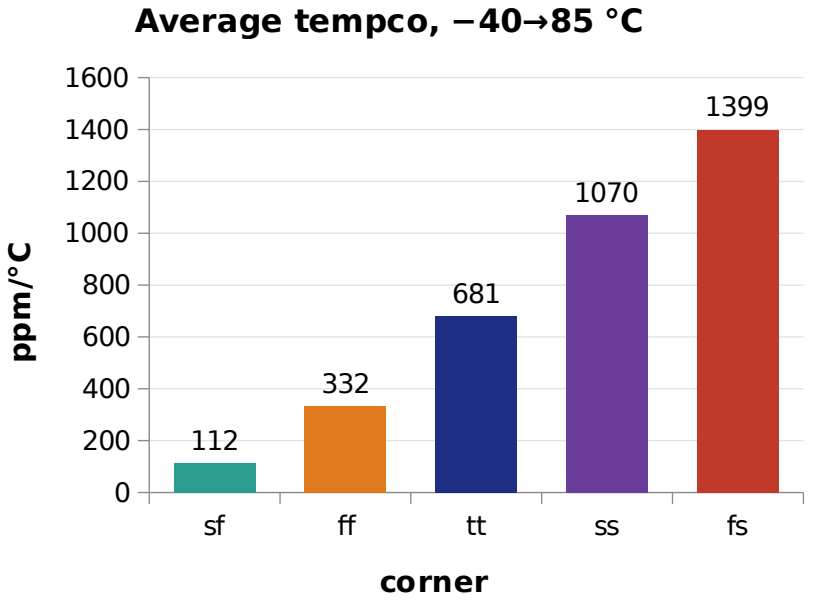
12.5x
spread, not the 3.4x previously reported

669
MHz/V K_VCO (tt, 0.9V, fit-derivative)

- Sensitivity insight: N/P-mismatched corners (sf, fs) bracket the entire tempco range, not the matched-speed corners (ff, ss) -- points at current-source balance, not absolute device speed, as the driving mechanism.

Uncompensated Drift: All 5 Corners



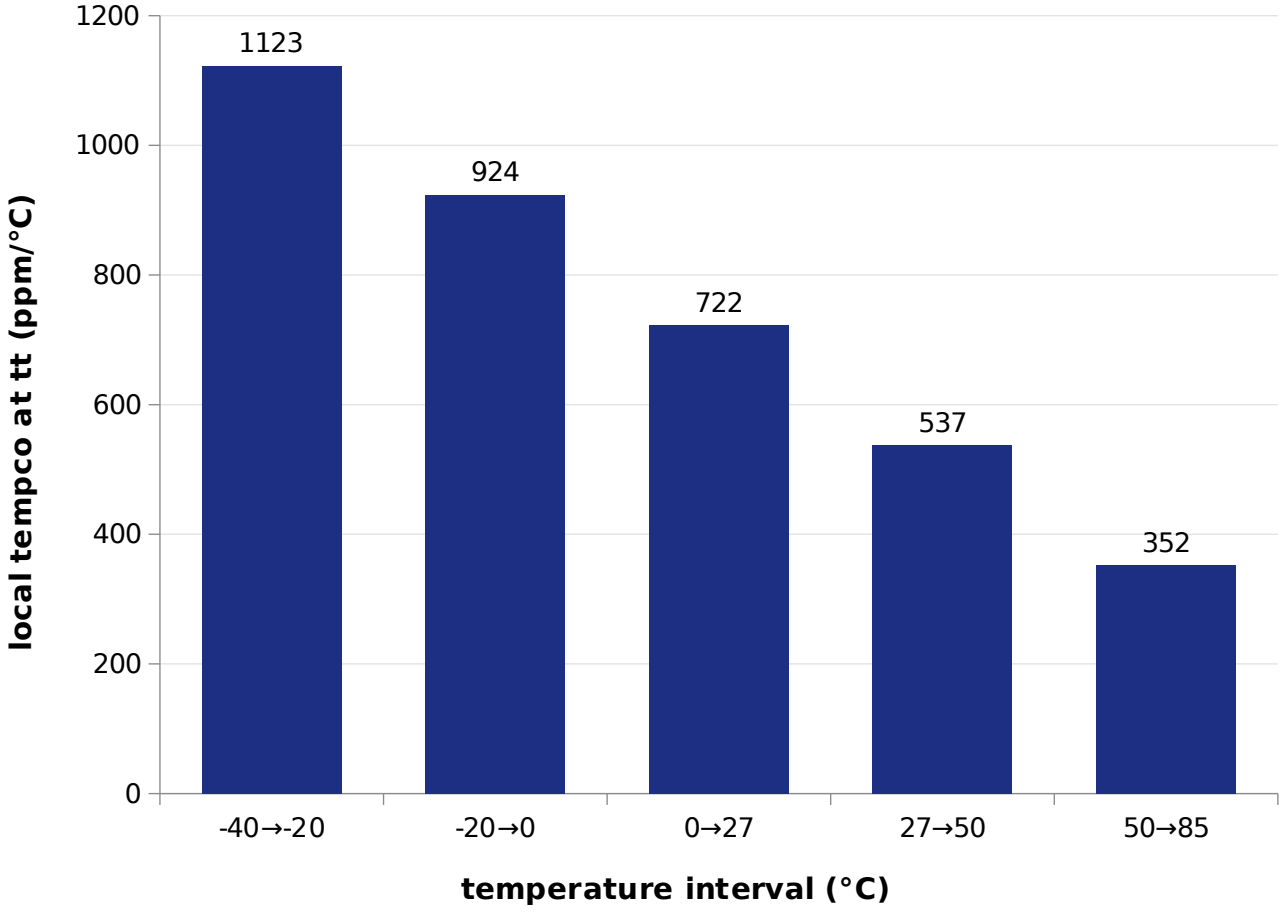


12.5×
tempco spread sf→fs

8.5 %
tt span, −40→85 °C

Every corner is far outside ±0.3 % uncompensated. N/P-mismatched corners (sf, fs) bracket the range.

Why Second Order? The Drift Is Curved



Reading the chart

If the drift were a straight line, every bar would be the same height (the 681 ppm/°C average).

Instead the slope falls 3.2× from cold (1123) to hot (352) — the drift is curved.

Consequence

A linear (1st-order) correction can only match the average slope: it leaves a bowed residual.

A quadratic term $k2 \cdot (T-27)^2$ matches the curvature → 2nd-order compensation.

Results & Analysis (contd.)

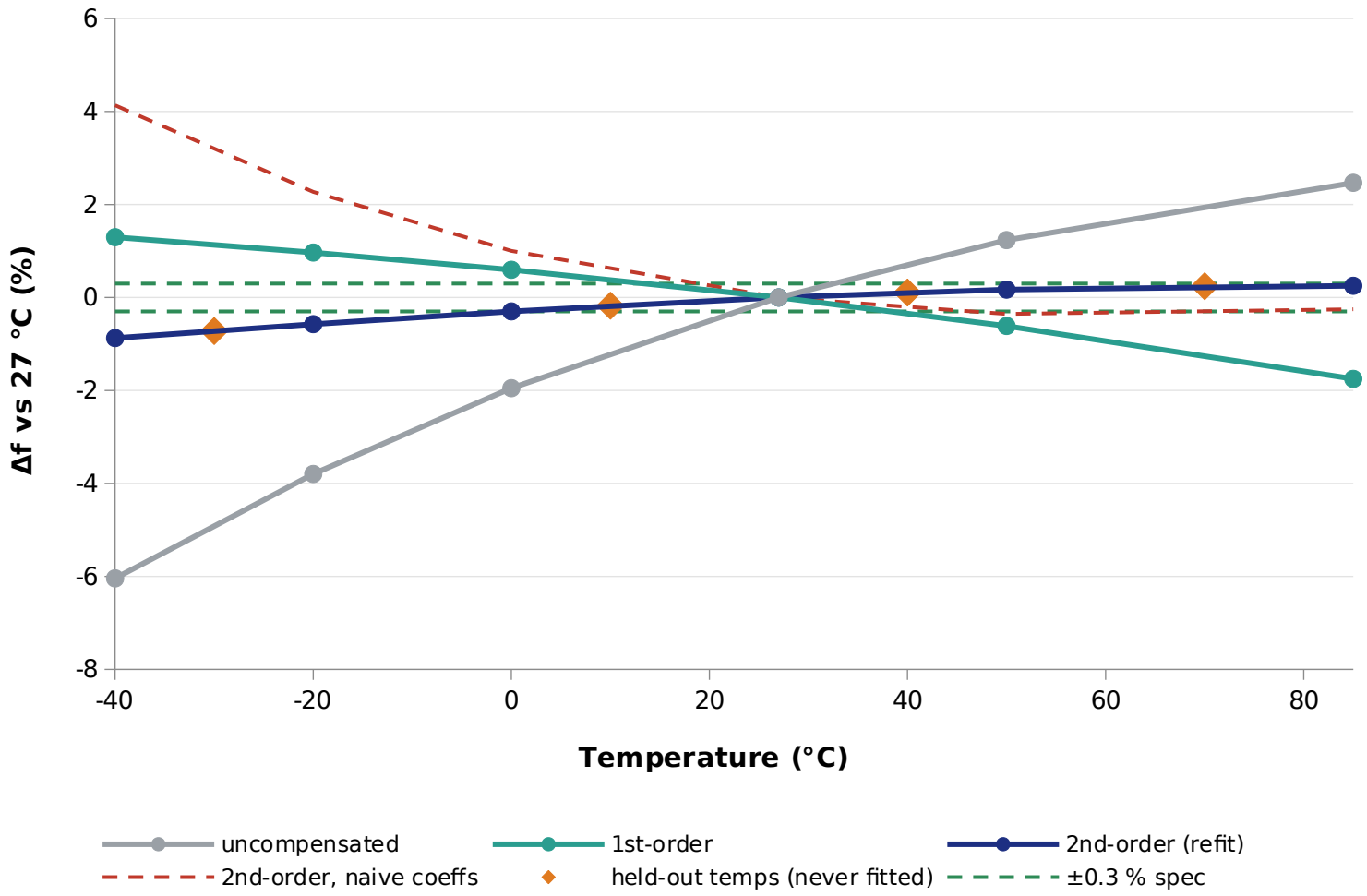
- Compensation refit independently at every corner; every corner improves at both stages. Checked at 4 held-out temperatures never used in fitting (-30/10/40/70C): all stay inside the fitted envelope.

Corner	Baseline span	1st-order span	2nd-order span	2nd-order max |dev|	Meets +/-0.3%?
ff	4.15%	1.06%	0.10%	0.07%	Yes
sf	1.48%	0.42%	0.24%	0.15%	Yes
tt	8.51%	3.05%	1.12%	0.87%	No
ss	13.37%	6.37%	3.32%	2.46%	No
fs	17.49%	9.13%	5.31%	3.83%	No

- **Our +/-0.3% target is met at ff and sf only -- not at the typical corner. Closing the gap needs a transistor-level network whose PTAT slope tracks process, and/or a per-corner trim (Review 3).**
- Naive analytical 2nd-order coefficients made tt worse (4.49% vs 3.05%): they assume constant K_VCO, which is false. Refitting against the measured 1st-order residual fixed it.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Compensation at the Typical (tt) Corner



8.51 → 1.12 %
span: baseline → 2nd-order

7.57×
compensation ratio C at tt

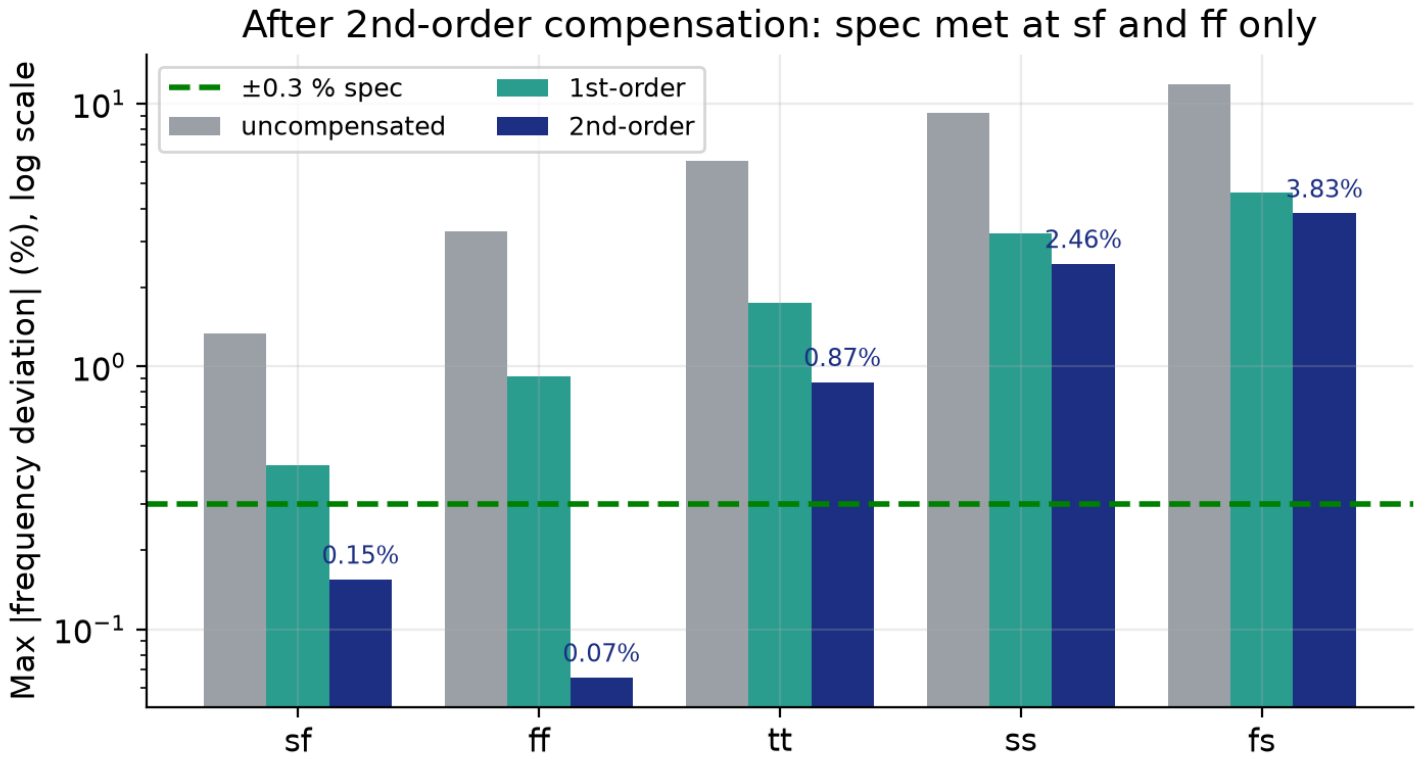
±0.87 %
max |dev|: spec ±0.3 % NOT met

Naive analytical k2 over-corrects the cold end (+4.1 % at −40 °C); refitting on the 1st-order residual fixes it.

Orange diamonds: temperatures never used in fitting, all inside the envelope.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Residual Drift per Corner vs ±0.3 % Spec



ff: 0.07 %
meets spec

sf: 0.15 %
meets spec

tt: 0.87 %
misses spec

ss: 2.46 %
misses spec

fs: 3.83 %
misses spec

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

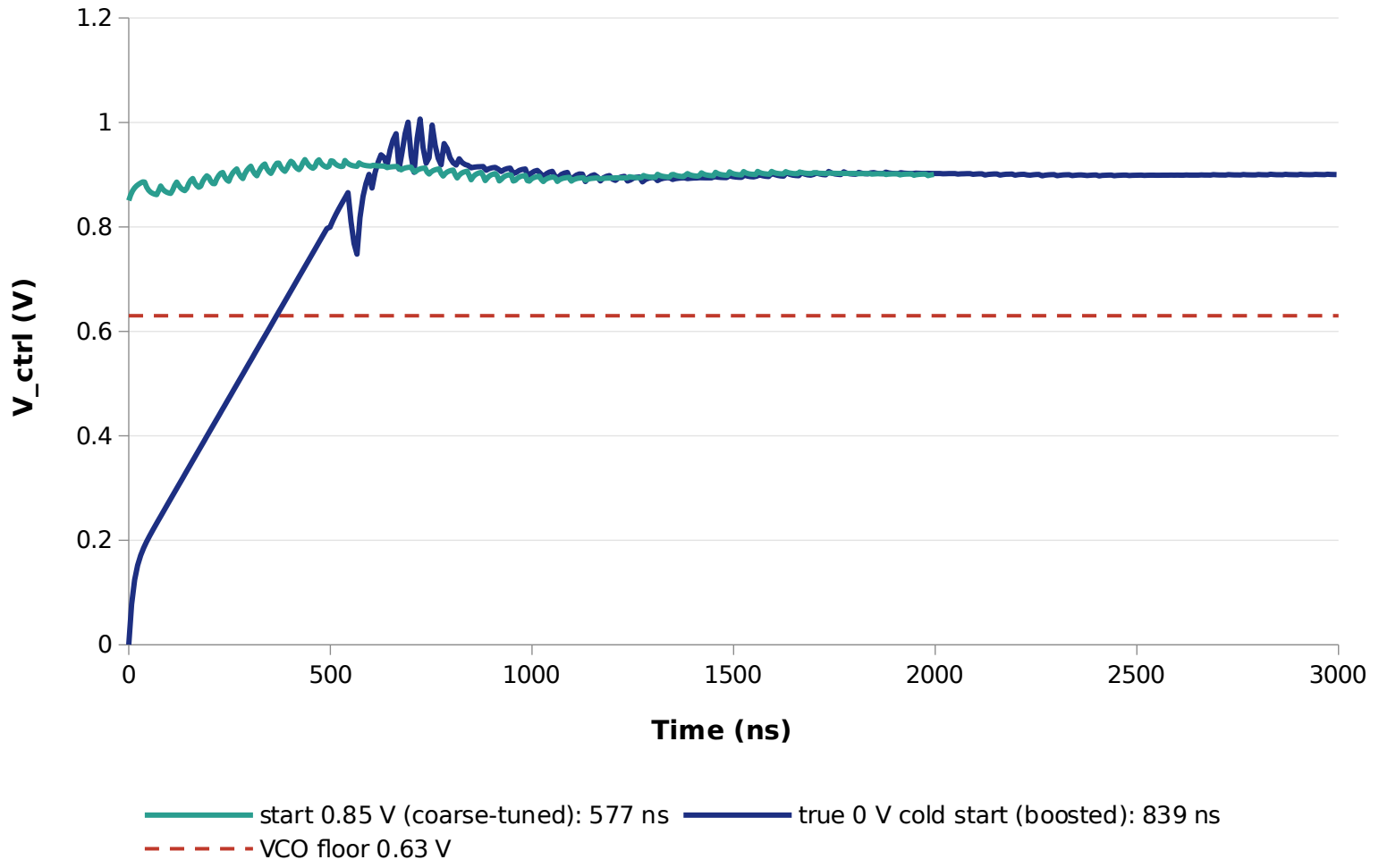
Results & Analysis (contd.) -- Closed-Loop Lock

Quantity	Coarse-tuned start (0.85V)	TRUE cold start (0V, boosted)
Lock time	577 ns	839 ns
Settled V_ctrl	0.902 V	0.9002 V
Settled f_VCO	227.96 MHz	226.42 MHz
Spec / prediction	< 2us spec; ~450ns analytical	< 2us spec

- Cold-start acquisition dead zone (open risk since Review 1) -- fixed and verified. Built a 5x-boosted-current charge pump variant that reverts to nominal current once the divider produces its first edge (tracked via a B-source-charged integrating capacitor, reusing the same current-injection idiom already proven correct -- not a new digital latch and its metastable-initial-state risk).
- Simulated from a TRUE 0V cold start (not the coarse-tuned 0.85V assumption): locks in 839 ns, comfortably inside the 2us spec -- worked on the first simulation attempt.
- Two real bugs found and fixed while closing the original loop (Review 1): an inverted SPICE B-source node-order convention (caught with a 5-line isolated test before touching the full loop), and an inconsistent loop-filter capacitor initial condition (a 50ns charge-pump hold gate was added to test a wrong hypothesis first -- disproving it is what pointed at the real cause).

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Closed-Loop Lock Transient



839 ns
lock from 0 V cold start

577 ns
lock from 0.85 V

< 2 µs
spec (analytic model: 450 ns)

Lock = V_ctrl within ±2 % of 0.9 V and staying there.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Results (contd.) -- Monte Carlo & Benchmark

Arm (tt, mismatch on, paired by seed)	n	Mean |drift|, 85C vs -40C	95% CI
Baseline (uncompensated)	12	18.93 MHz (8.4%)	[16.20, 21.65]
1st-order	30	5.85 MHz (2.6%)	[4.62, 7.09]
2nd-order (curvature-matched)	30	3.37 MHz (1.5%)	[2.33, 4.41]

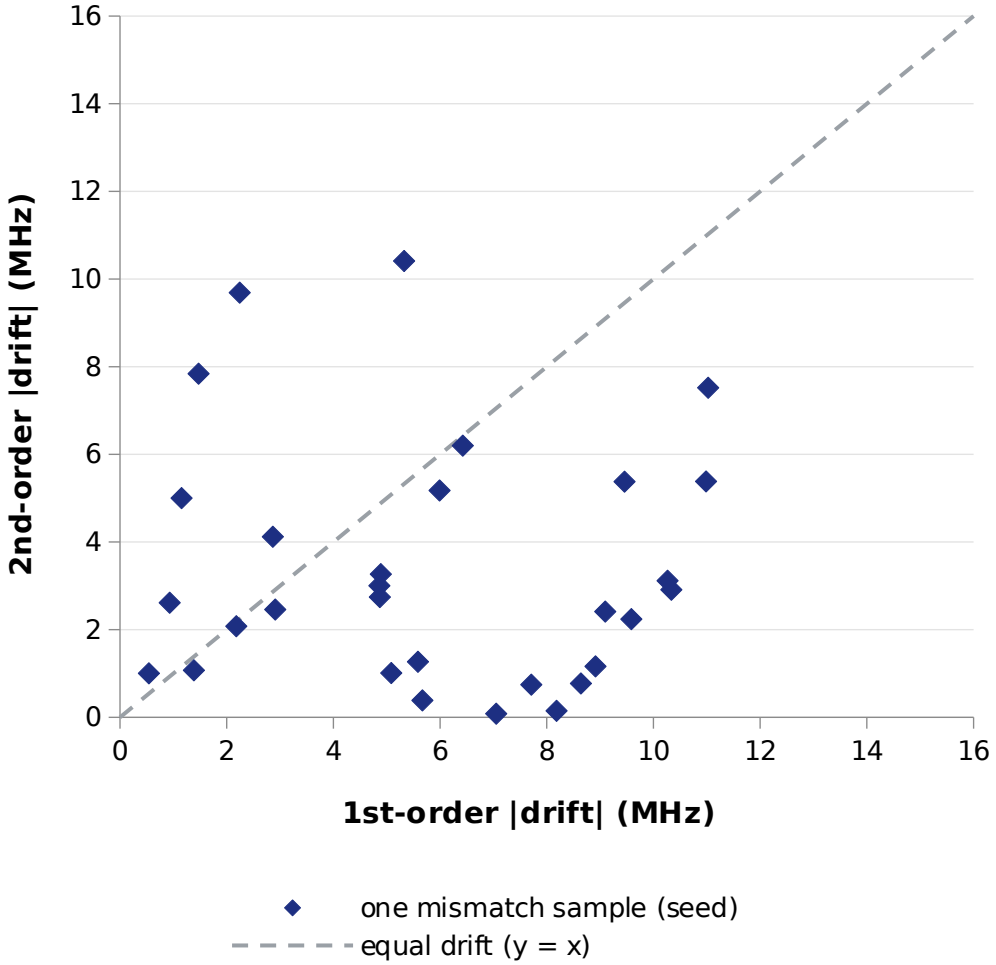
- **Both stages significantly reduce drift (12/12 seeds). 2nd-order beats 1st-order: -2.48 MHz, 95% CI [0.81, 4.15], paired t-test and Wilcoxon agree, 23/30 seeds.**
- Correction on record: our first pass tested signed drift and overstated this at n=12 (1st-order over-corrects, so the sign flipped). On |drift| it was not significant at n=12 -- n=30 is what earns the claim.

Design	Process	Residual drift	Lock time
This work, 2nd-order (tt)	sky130, simulation	+/-0.87% max dev., -40 to 85C	839 ns (0V cold start)
Souliotis et al. 2022 [1]	180nm, measured silicon	-1.5% / +0.4%, -40 to 120C	not reported
Stancu et al. 2023 [2]	simulation, 50 MHz clock	1.6% temperature error	n/a
Geng et al. 2018 [3]	180nm, measured silicon	not reported as drift	< 2 us

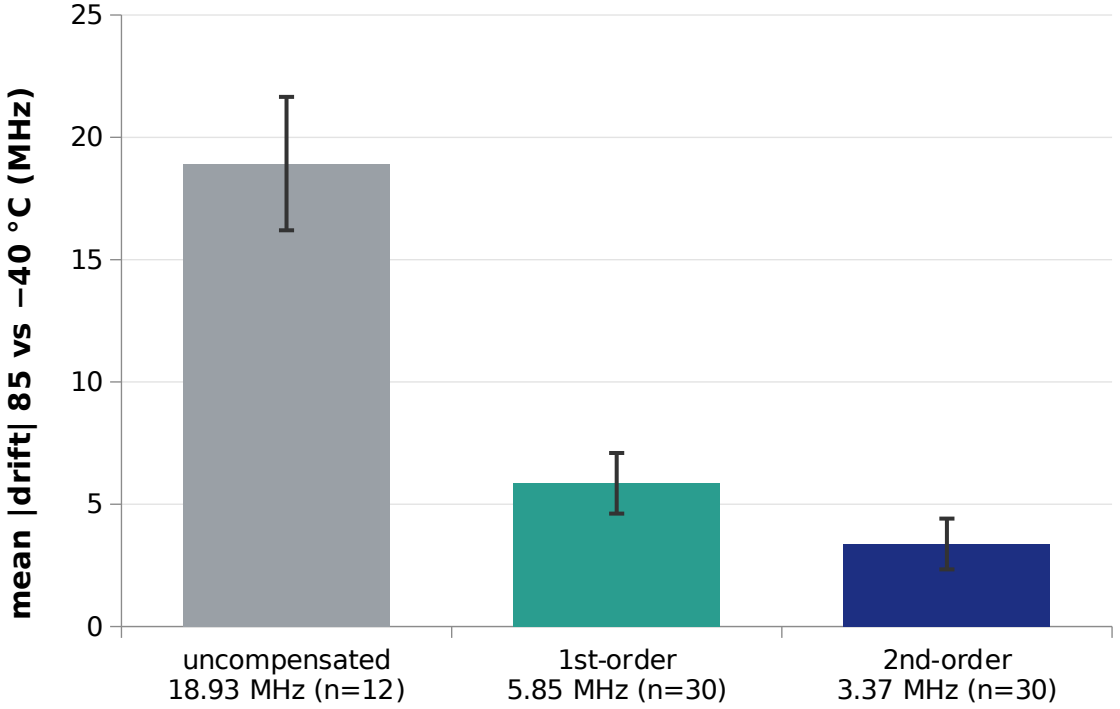
- Comparable to published work, but ours is simulation of an ideal correction network vs. their measured silicon -- not a like-for-like claim until the transistor-level network exists.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Monte Carlo: Paired Comparison (tt)



Points below the dashed line: 2nd-order beat 1st-order on that same chip.



23 / 30
seeds where 2nd < 1st

−2.48 MHz
95 % CI [0.81, 4.15]

p < 0.05
paired t + Wilcoxon

Review 2 at a Glance

12.5×
tempco spread across 5 corners (112→1399 ppm/°C)

669 MHz/V
K_VCO at 0.9 V, tt

1.12 %
tt span after 2nd-order (was 8.51 %)

2 / 5
corners meeting ±0.3 % (ff, sf)

3.37 MHz
Monte Carlo mean |drift| after 2nd-order (was 18.93)

23 / 30
seeds where 2nd-order beats 1st

839 ns
lock from a true 0 V cold start (< 2 µs)

15
IEEE references; 1 patent narrowed our novelty claim

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Challenges & Remaining Work

- **Challenges faced and solutions**
  - SPICE B-source polarity and an inconsistent loop-filter initial condition broke the first closed-loop runs -- each isolated with a minimal test (and one wrong hypothesis disproved) before touching the full loop.
  - Naive analytical 2nd-order coefficients over-corrected the cold end -- fixed by refitting against the measured 1st-order residual; the refit was then checked at 4 held-out temperatures it never saw.
  - Caught our own statistics error: the first Monte Carlo pass compared SIGNED drift, which made 1st- vs 2nd-order look significant only because the sign flipped. Re-tested on |drift| and re-run at n=30.
  - Novelty check found a 2017 patent on generic quadratic compensation -- narrowed our claim to ring-VCO + closed-loop PLL + open PDK + statistically validated quantification.
- **Remaining work (25%)**
  - Close the spec gap: +/-0.3% is met at ff and sf only (tt 0.87%, ss 2.46%, fs 3.83%) -- a transistor-level compensation network whose PTAT slope tracks process, plus a per-corner trim.
  - Transistor-level charge pump; layout in Magic, DRC, LVS (netgen), post-layout re-simulation.
  - Cadence/Virtuoso cross-validation arm and the actual H1 regression (eps_res vs C) -- the other half of the research hypothesis, pending lab access.
- **Timeline to Review-III (28.10.2026)**
  - Weeks 1-2: transistor-level CP + compensation network. Week 3: layout / DRC / LVS. Week 4: Virtuoso arm, H1 regression, final report.

References

- [1] G. Souliotis, E. Keramida, F. Plessas, R. Malatesta, and S. Vlassis, “Temperature compensated ring oscillator based VCO,” AEU – Int. J. Electron. Commun., vol. 149, Art. no. 154195, May 2022.
- [2] C. Stancu, A. Neacsu, O. Profirescu, D. Dobrescu, and L. Dobrescu, “Temperature and power supply compensated CMOS clock circuit based on ring oscillator,” Electronics, vol. 12, no. 3, Art. no. 507, Jan. 2023.
- [3] Z. Geng, X. Ma, and Z. Li, “A low power fast locking PLL frequency synthesizer with temperature compensation,” in Proc. 3rd Int. Conf. EAME, Atlantis Press, Jun. 2018, pp. 281–285.
- [4] Y. You, D. Huang, J. Chen, and S. Chakraborty, “A 12GHz 67% tuning range 0.37ps RJrms PLL with LC-VCO temperature compensation scheme in 0.13µm CMOS,” in Proc. IEEE RFIC, Tampa, FL, USA, Jun. 2014, pp. 101–104.
- [5] S. Sinha, “Design of an integrated CMOS PLL frequency synthesizer,” in Proc. 11th IEEE MELECON, Cairo, Egypt, 2002, pp. 220–224.
- [6] É. M. V. Freitas et al., “Using open source EDA tools in ASICs for HEP: A mixed comparison,” J. Instrum., vol. 20, no. 12, Art. no. P12022, Dec. 2025.
- [7] M. Thiriloganathan et al., “A 2.4 GHz LC-VCO fractional-N phase locked loop open-source design in 130-nm BiCMOS,” in Proc. 22nd Int. Conf. SMACD, 2026, arXiv:2607.08852.
- [8] R. N. S. Raphael, A. M. Pinto, L. T. Manera, and S. Finco, “Phase-locked loop (PLL)-based frequency synthesizer for digital systems driving,” in Proc. BTSym’18, SIST vol. 140, Springer, 2019, pp. 395–406.
- [9] Y. Li, K. Du, and J. Zhang, “Design of ring oscillator with temperature compensation effect,” Int. J. RF Microw. Comput.-Aided Eng., vol. 32, no. 11, Art. no. e23374, Nov. 2022.
- [10] H. H. Hammam, K. M. Hassan, and S. A. Ibrahim, “An ultra-low-power process-and-temperature compensated ring oscillator,” in Proc. 9th Int. Conf. ICEEE, Alanya, Turkey, Mar. 2022.
- [11] S. Zhang, A. Li, Y. Han, L. Jie, X. Han, and R. C. C. Cheung, “Temperature compensation technique for ring oscillators with tail current,” Electron. Lett., vol. 52, no. 13, pp. 1108–1110, Jun. 2016.
- [12] M. Ortner and M. Punzenberger, “Systems and methods for oscillators using quadratic temperature compensation,” U.S. Patent 9 748 962 B2, Aug. 29, 2017.
- [13] D. Patrinos and G. Souliotis, “A low-voltage, low-power 2.5 GHz ring oscillator with process and temperature compensation,” J. Low Power Electron. Appl., vol. 15, no. 3, Art. no. 52, Sep. 2025.
- [14] M. Sivasakthi and P. Radhika, “Design and analysis of PVT tolerant hybrid current starved ring VCO with bulk driven keeper technique at 45 nm CMOS technology for the PLL application,” AEU – Int. J. Electron. Commun., vol. 173, Art. no. 154987, Jan. 2024.
- [15] S. Madheswaran and R. Panneerselvam, “Design and analysis of 7-stage MOS current mode logic power gated MOSFETs in current starved VCO for the PLL application,” Int. J. Electr. Comput. Eng., vol. 14, no. 2, pp. 1398–1405, Apr. 2024.

Thank You

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